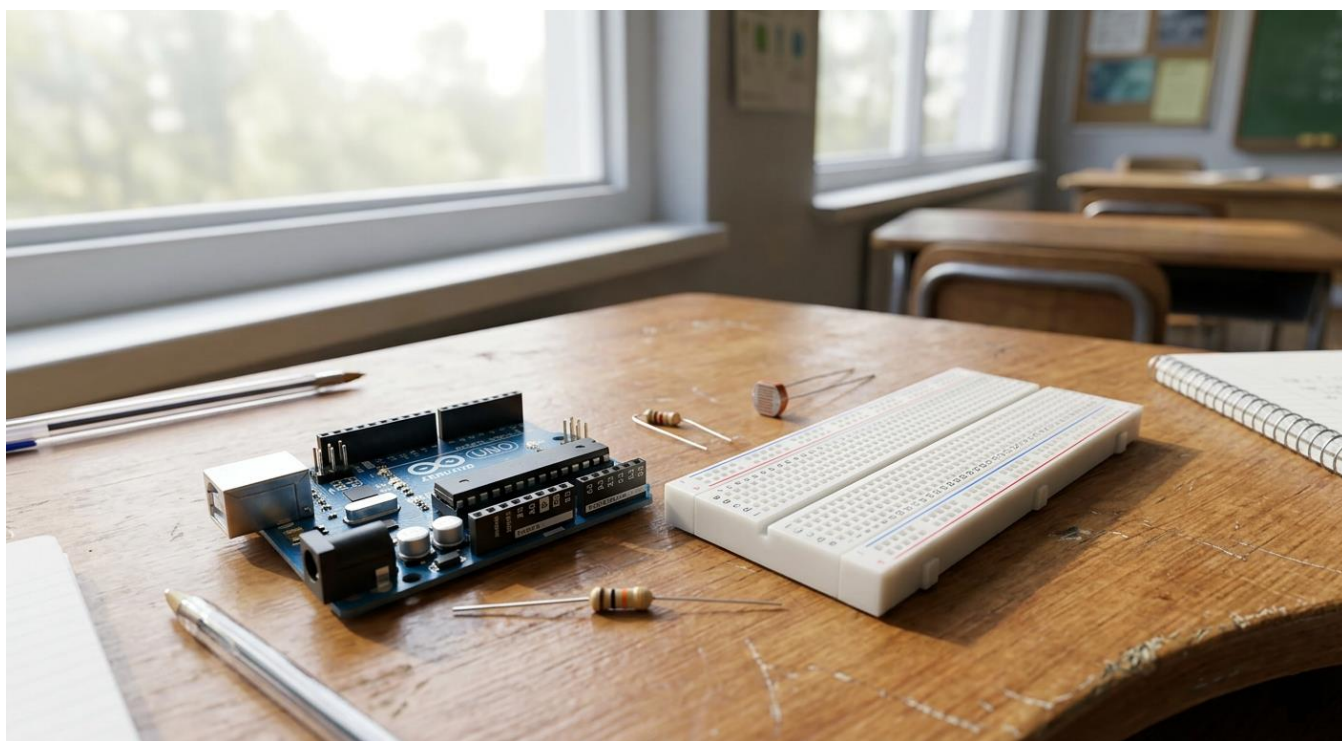


⚡ Math-O-Tronics · STEM + Math Integration Resources

💡 Arduino Light Sensor: Fractions & Ratios Lab

Collect Real Sensor Data · Apply Real Math · Build Real Understanding

GRADES	DURATION	SUBJECT	PREP TIME	PAGES
6 – 7	3 – 4 Days	Math + STEM	10 – 15 min	16



✓ Hands-On STEM Math	✓ Arduino Ready	✓ No Hardware Option
✓ Print + Digital Ready	✓ Grades 6 – 7	✓ Answer Key Included

🌀 CCSS 6.RP.A.1 & 6.RP.A.3 | NGSS Practice 3 | ISTE Student Standard 5



Teacher Resource Overview

Resource Overview

In this lab, students use an Arduino microcontroller and a photoresistor (LDR) to measure light intensity in different classroom locations. Raw sensor values (0–1023) are first recorded, then converted to a percentage of maximum light. Students use these real numbers to write fractions, simplify them, compare ratios, and express proportional relationships — making CCSS 6.RP.A concrete and meaningful.



Standards Alignment

Area	Standard Code	Focus
Common Core Math	CCSS 6.RP.A.1	Understand the concept of a ratio; use ratio language to describe a relationship
Common Core Math	CCSS 6.RP.A.3	Use ratio and rate reasoning to solve real-world problems
Common Core Math	CCSS 6.NS.B.2 / 6.RP.A	Convert raw values to percentages; compare fractions
NGSS	Practice 3	Planning and carrying out investigations; collecting sensor data
ISTE Student	Standard 5	Computational thinker — analysing data collected with technology



Learning Objectives (first 3 of 6)

1	Collect raw light sensor readings (0–1023) using an Arduino + LDR circuit.
2	Convert raw values to a percentage of maximum light using the formula: $\% = (\text{raw} \div 1023) \times 100$.
3	Write the light level of one location as a fraction of another location's reading.

[+ 3 more learning objectives in the full version]

Part A — Build Your Light Sensor Circuit

GOAL: Wire the photoresistor (LDR) to your Arduino, upload the starter code, and verify that the Serial Monitor shows changing numbers when you cover or uncover the sensor.

How the Photoresistor Works

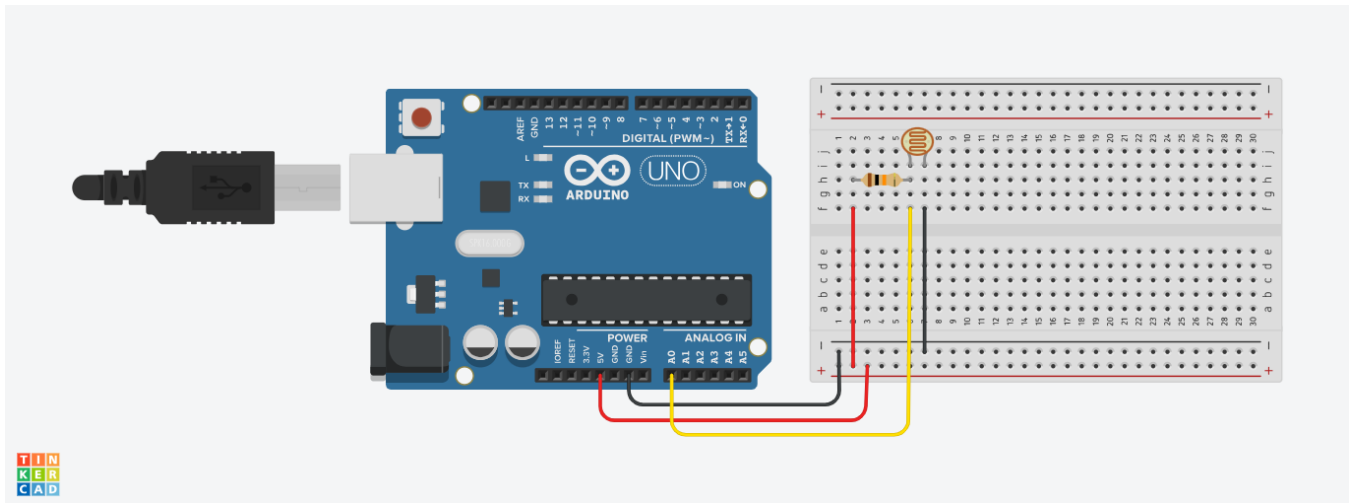
A photoresistor (LDR) changes its electrical resistance depending on light:

More light → lower resistance → higher number in the Serial Monitor

Less light → higher resistance → lower number in the Serial Monitor

Reading range: 0 (total darkness) → 1023 (maximum brightness)

Wiring Diagram



Wiring Reference Table

Component / Pin	Goes To	Arduino Connection	Wire Colour
10 kΩ resistor — Leg 1	Any breadboard row	5V (power)	Red
10 kΩ resistor — Leg 2	Same row as LDR Leg 1	Junction → A0	Yellow
LDR — Leg 1	Junction with resistor	A0 (Analog In)	Yellow
LDR — Leg 2	GND rail	GND	Black

Part C — Fractions, Ratios & Analysis

GOAL: Use your percentage values from Part B to write fractions comparing locations, simplify them, and express comparisons as ratios. Then answer the analysis questions.

☆ STEP 0 — Rounding Rule (Do This First!)

Before writing any fraction, round your Light Level (%) to the nearest whole number. This way you avoid trying to simplify messy decimals like 86.7 / 54.7.

For example: **86.7% → 87** | **54.7% → 55** | **19.6% → 20**

Now you can work with whole numbers and simplify cleanly!

Section 1 — Writing & Simplifying Fractions

A fraction can compare the light level of one location to another. Write the numerator (top) and denominator (bottom), then simplify as far as possible.

Fraction 1 — Location A compared to Location B		
Write the fraction	Simplify (show steps)	Simplest form
A = _____ % B = _____ % Fraction: _____ / _____	 _____ / _____	 _____ / _____

[Full version includes Fraction 2, Fraction 3, Ratio Tables, 5 Analysis Questions + Extension Challenge]